Ex : 10
Date : 27/08/26

Internetworking with routing in CISCO Packet Tracer Simulator.

A.) Design and Configure a simple Internetwork router in CISCO Packet Tracer :-

AIM :-

To design & configure two different network using a router and switches, assign static IP address and verify packet communication between the networks.

Steps :-

1.) Open CISCO Packet Tracer.
2.) Add 1 router, 2 switches and 3-4 pcs.
3.) Connect Pcs to switches using copper straight-through Cables.
4.) Connect both switches to the router.
5.) Assign static IP address to all Pcs.
6.) Configure the router interfaces with suitable IP address
7.) Enable the router interface using no shutdown.
8.) Set the router interface IP as the default gateway for each network.
9.) Use the ping command to check Connectivity.
10.) Select simulation mode and observe the packet transmission path through the router.

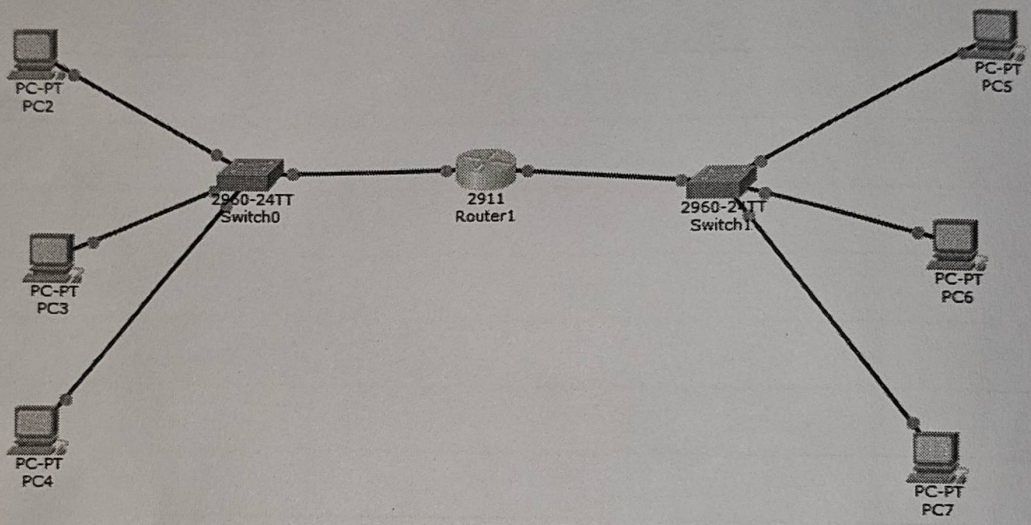
RESULT :-

The two network were successfully connected using a router, and communication between different networks was verified successfully using the ping command and simulation mode.

```
Router(config)#interface gigabitethernet 0/1
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state
to up
exit
```




```
Router#enable
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.10.1    YES manual up                    up
GigabitEthernet0/1     192.168.20.1    YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

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10.b.)
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Inter network using wireless router DHCP Server and Internet could

AIM :-

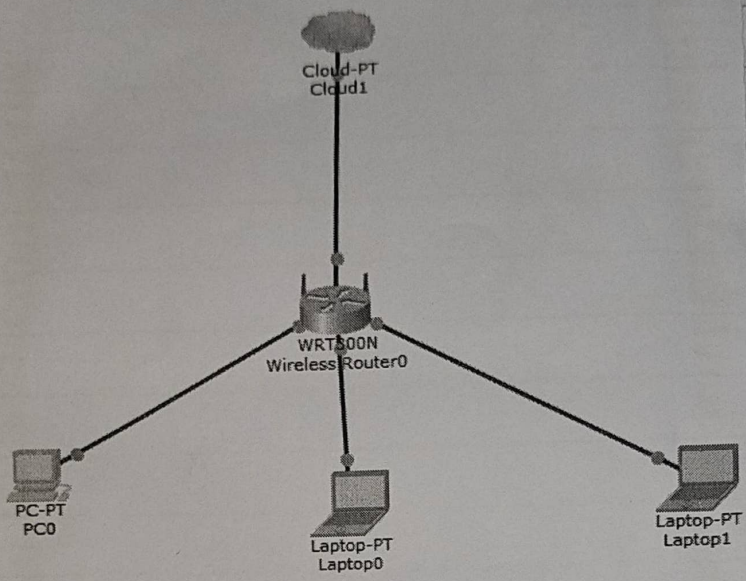
To design and configure a wireless internetwork using a wireless router, DHCP server and internet cloud and verify automatic IP address allocation & connectivity.

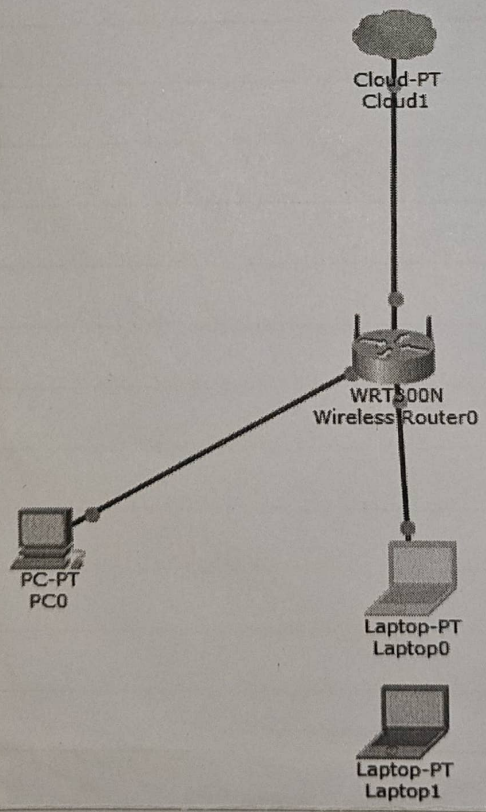
Steps :-

1.) Open CISCO Packet Tracer
2.) Add a wireless Router, Laptops / wireless devices and an internet cloud.
3.) Connect the wireless router to the internet cloud using an ethernet cable.
4.) Configure the SSID of the wireless router.
5.) Enable the DHCP server on the wireless router.
6.) Configure the DHCP IP address range.
7.) Connect Laptops to the wireless network
8.) Set the laptops to obtain IP address automatically using DHCP
9.) Check whether the devices receive IP address automatically.
10.) Use the ping command to test connectivity.
11.) Use simulation mode to observe packet transmission.

RESULT :-

The wireless internetwork was successfully configured using a wireless router DHCP server and Internet cloud. IP addresses were automatically assigned to wireless devices and connectivity was successfully verified.


Cloud-PT
Cloud1
WRT300N
Wireless Router0
PC-PT
PC0
Laptop-PT
Laptop0
Laptop-PT
Laptop1


Cloud-PT
Cloud1
WRT300N
Wireless Router0
PC-PT
PC0
Laptop-PT
Laptop0
Laptop-PT
Laptop1

10.(i) Internetwork using Switch, Router and Ethernet cable.

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AIM:-

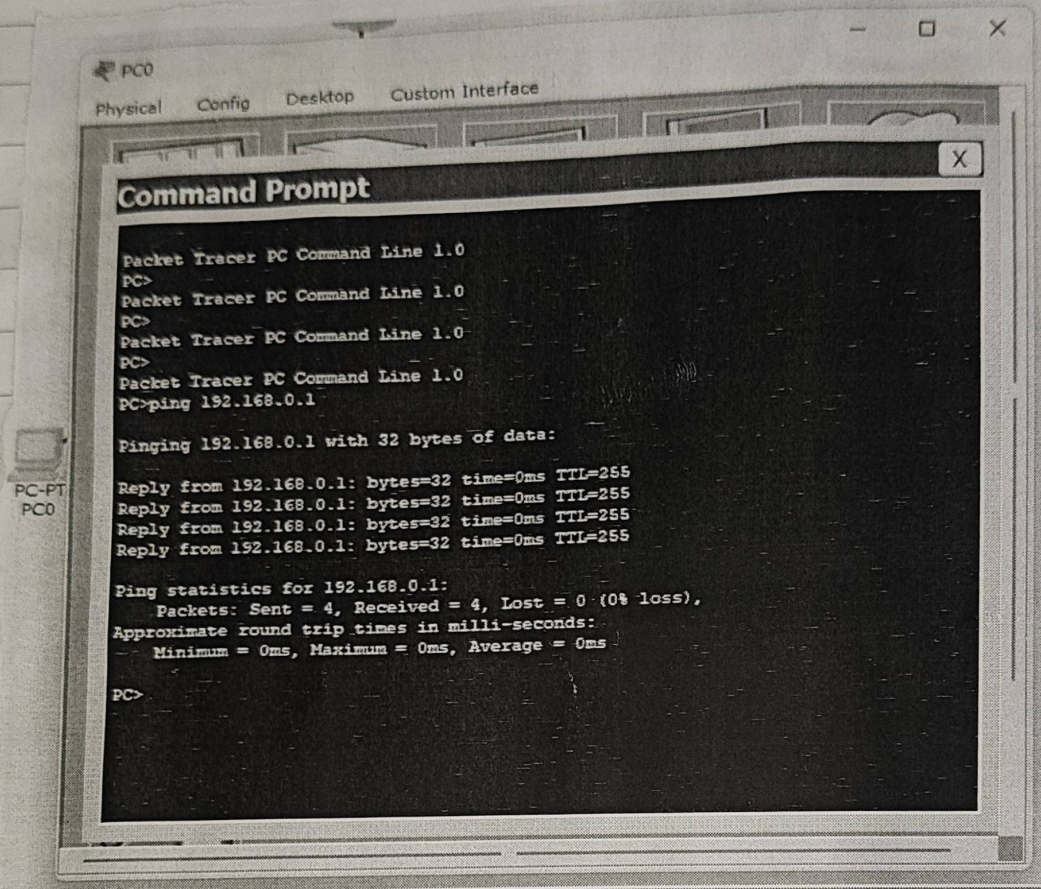
To design and configure an internetworking using switches, a router and ethernet cables and to establish communication between different networks.

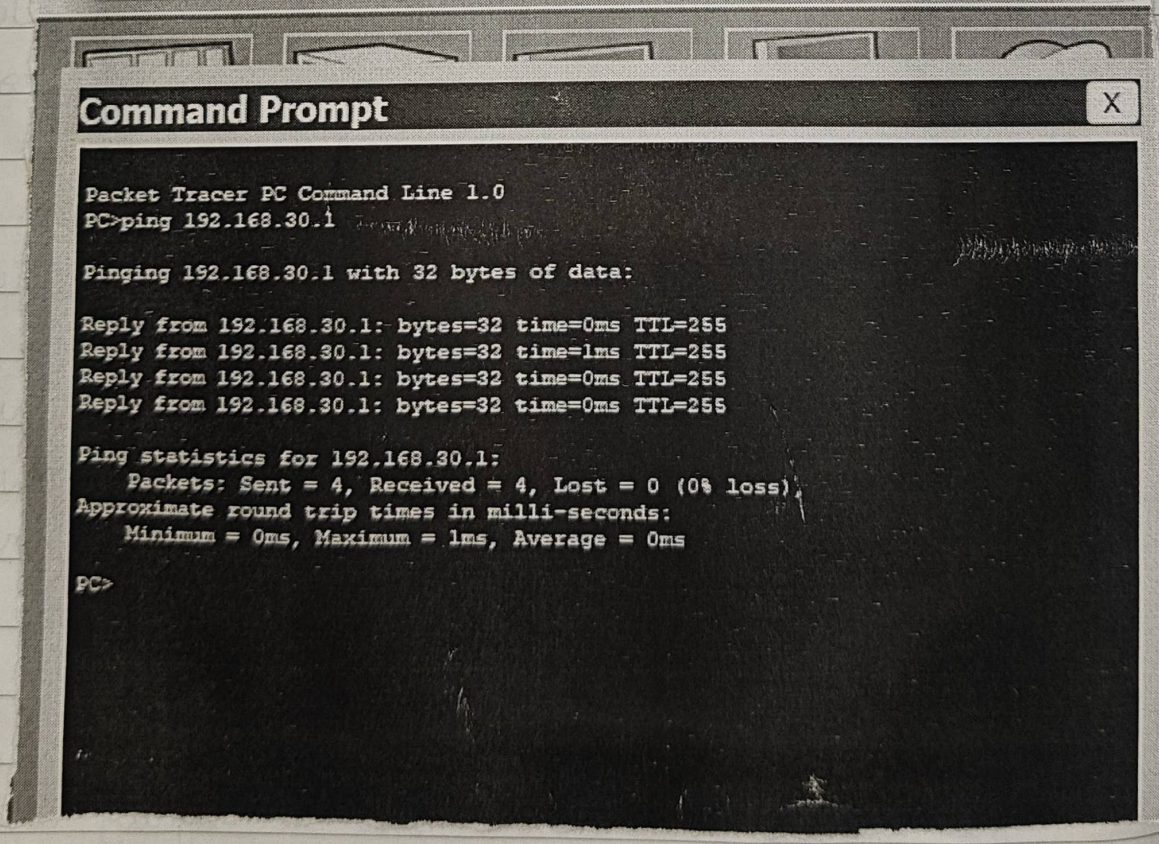
Steps:-

1) Open CISCO packet Tracer
2) Add 1 router, 2 switches, and 3-4 PCs.
3) Connect the PCs to the switches using Ethernet cables.
4) Connect the switches to the router using ethernet cables.
5) Assign suitable static IP addresses to the PC's.
6) Configure the IP address of the router Interfaces.
7) Enable the router interfaces using no shutdown.
8) Configure the appropriate default gateway on each PC.
9) Use the Ping command to test communication between PC.
10) Select & Simulation Mode and observe the packet flows through the switches and router.

RESULT:-

The internetwork was successfully designed and configured using switches and router and ethernet cable Communication between the different network was established and verified successfully using the ping command.


PC0
Physical
Config
Desktop
Custom Interface
Command Prompt
Packet Tracer PC Command Line 1.0
PC>
Packet Tracer PC Command Line 1.0
PC>
Packet Tracer PC Command Line 1.0
PC>
Packet Tracer PC Command Line 1.0
PC>ping 192.168.0.1
Pinging 192.168.0.1 with 32 bytes of data:
Reply from 192.168.0.1: bytes=32 time=0ms TTL=255
Reply from 192.168.0.1: bytes=32 time=0ms TTL=255
Reply from 192.168.0.1: bytes=32 time=0ms TTL=255
Reply from 192.168.0.1: bytes=32 time=0ms TTL=255
Ping statistics for 192.168.0.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
PC>
PC-PT
PC0


Command Prompt
Packet Tracer PC Command Line 1.0
PC>ping 192.168.30.1
Pinging 192.168.30.1 with 32 bytes of data:
Reply from 192.168.30.1: bytes=32 time=0ms TTL=255
Reply from 192.168.30.1: bytes=32 time=1ms TTL=255
Reply from 192.168.30.1: bytes=32 time=0ms TTL=255
Reply from 192.168.30.1: bytes=32 time=0ms TTL=255
Ping statistics for 192.168.30.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
PC>